# DEVANSH AWASTHI

**Graduate Engineer Trainee | DevOps & Cloud Engineer | Platform Engineering**

Azure DevOps • Kubernetes • Terraform • Docker • CI/CD Automation • Infrastructure as Code

## PROFESSIONAL SUMMARY


Results-driven Graduate Engineer Trainee specializing in DevOps Engineering, Cloud Infrastructure, and Platform Engineering with hands-on experience designing and delivering enterprise-grade DevOps solutions on Microsoft Azure. Demonstrated expertise in end-to-end CI/CD pipeline automation, Infrastructure as Code (IaC), container orchestration, configuration management, and monitoring & observability across cloud-native environments.

Proficient in Terraform-based infrastructure provisioning, Ansible-driven configuration management, Docker containerization, Kubernetes cluster administration, Azure DevOps pipeline engineering, JFrog Artifactory artifact management, SonarQube DevSecOps integration, and Prometheus/Grafana monitoring. Experienced in applying Shift Left practices, GitOps principles, and Agile SDLC methodologies to accelerate release cycles, improve deployment reliability, and drive operational excellence.

Additionally, brings an AI/ML research background with a proven 97% model accuracy achievement in computer vision projects, enabling a unique perspective on AI-augmented DevOps automation and intelligent platform engineering.


## TECHNICAL SKILL MATRIX

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|---|---|
| **Cloud Platforms** | Microsoft Azure (VM, VNet, Storage, Resource Groups, Azure DevOps) |
| **DevOps Tools** | Azure DevOps, Git, GitHub, SonarQube (DevSecOps), GitHub Copilot |
| **CI/CD Automation** | Azure DevOps Pipelines, CI/CD Design, Release Automation, Deployment Automation, GitOps |
| **Container Tech** | Docker, Kubernetes (K8s), Container Orchestration, Microservices Deployment, Cluster Management |
| **IaC & Automation** | Terraform, Ansible, Infrastructure Provisioning, Configuration Management, Shell Scripting |
| **Monitoring & Obs.** | Prometheus, Grafana, Monitoring & Alerting, Dashboarding, Observability, Incident Troubleshooting |
| **Artifact Management** | JFrog Artifactory, Maven, Version Control, Artifact Lifecycle Management |
| **Programming** | Python, Java, Bash/Shell Scripting, PowerShell |
| **Operating Systems** | Linux (Ubuntu/RHEL), Windows Server |
| **Methodologies** | Agile, SDLC, DevSecOps, Shift Left, High Availability, Cloud-Native Architecture |

## PROFESSIONAL EXPERIENCE

### Graduate Engineer Trainee – DevOps & Cloud

*March 2026 – June 2026*

**Self-Managed DevOps & Kubernetes Platform | Internal Enterprise DevOps Transformation | Microsoft Azure**

Designed and implemented a production-grade, end-to-end DevOps platform on Microsoft Azure encompassing infrastructure provisioning, configuration management, containerization, Kubernetes orchestration, CI/CD pipeline automation, artifact management, and monitoring. The platform delivers repeatable, scalable, and highly observable deployments for containerized microservices.

**Key Responsibilities & Achievements:**

- Provisioned and managed Azure Virtual Machine infrastructure using Terraform Infrastructure as Code (IaC), ensuring repeatable, version-controlled, and scalable cloud resource deployments — reducing manual provisioning effort and improving environment consistency.

- Automated end-to-end server configuration and software installation across cloud VMs using Ansible playbooks, eliminating configuration drift and reducing setup time for new environments.
- Configured Linux (Ubuntu) environments for application deployment, Kubernetes node preparation, and platform operations — ensuring secure, optimized, and production-ready infrastructure.
- Architected and administered a multi-node Kubernetes cluster for container orchestration, enabling high-availability microservices deployment, workload scheduling, and cluster lifecycle management.
- Containerized microservice applications using Docker, optimizing container image layers and enforcing image build best practices to enhance portability, security, and deployment efficiency.
- Designed and implemented multi-stage CI/CD pipelines in Azure DevOps covering automated build, static code analysis, artifact publishing, and Kubernetes deployment workflows — accelerating release cycles and improving deployment reliability.
- Integrated SonarQube for continuous code quality analysis and quality gate enforcement, embedding DevSecOps and Shift Left security practices into the CI/CD pipeline to detect vulnerabilities early.
- Configured JFrog Artifactory as a centralized artifact repository for managing build artifacts, container images, and Maven packages — establishing version-controlled artifact lifecycle management.
- Implemented automated Kubernetes deployment workflows for containerized applications, enabling consistent and repeatable application rollouts with minimal manual intervention.
- Established comprehensive monitoring and observability using Prometheus for metrics collection and Grafana dashboards for real-time visibility into infrastructure health, application performance, and alerting.
- Performed root cause analysis and incident troubleshooting across infrastructure, deployment pipelines, networking, and Kubernetes cluster issues — improving platform reliability and operational uptime.
- Applied GitOps principles and Infrastructure as Code standards to ensure all platform configurations are version-controlled, peer-reviewable, and auditable.
- Leveraged GitHub Copilot to accelerate automation scripting, pipeline development, and code review processes — improving developer productivity across platform engineering tasks.

**AI Research Engineer (Academic Project)** *January 2026 – April 2026*

**Flood ResQ – AI-Powered Human Detection in Flood Zones | Academic Research Project**

Developed a two-stage AI-powered flood detection and human localization system to support disaster response operations. Leveraged DenseNet and YOLOv8 deep learning models to identify flood conditions and detect stranded individuals in real-world affected regions.

- Designed and trained DenseNet-based deep learning models for binary flood detection, optimizing for precision and recall across diverse environmental conditions.
- Implemented real-time human localization using YOLOv8 object detection, enabling accurate identification of stranded individuals in flood-impacted imagery.
- Built scalable image preprocessing pipelines utilizing CLAHE contrast enhancement, normalization, and resizing techniques to improve model training quality across 10,000+ annotated images.
- Achieved 97% detection accuracy during model evaluation, demonstrating production-grade model performance suitable for real-world disaster management deployments.
- Optimized model performance and inference speed to meet real-world operational requirements for emergency response applications.

## EDUCATION

**Bachelor of Technology – Artificial Intelligence & Machine Learning** *Graduated 2026*

*Pranveer Singh Institute of Technology (AKTU), Kanpur, Uttar Pradesh, India*

CGPA: 7.28 / 10

## CERTIFICATIONS & TRAINING

- Oracle Cloud Infrastructure 2025 AI Foundations Associate
- Oracle Java Foundations
- Deep Learning Fundamentals – IBM
- Machine Learning with Python – IBM
- Generative AI Explained – NVIDIA
- Goldman Sachs Governance Analyst Virtual Experience Program

## CORE COMPETENCIES

DevOps Engineering | Cloud Engineering | Platform Engineering | Site Reliability Engineering (SRE) | Infrastructure Engineering | Azure DevOps Engineer | Kubernetes Engineer | Automation Engineer | CI/CD Pipeline Automation | Continuous Integration | Continuous Delivery | Continuous Deployment | Infrastructure as Code (IaC) | Configuration Management | Release Automation | Deployment Automation | Containerization | Kubernetes Administration | Cluster Management | Microservices Deployment | Artifact Management | DevSecOps | Shift Left Practices | Platform Reliability | High Availability | Monitoring & Alerting | Observability | Incident Troubleshooting | Root Cause Analysis | Change Management | Agile Methodology | SDLC | GitOps | Cloud-Native Architecture

## ACHIEVEMENTS & ADDITIONAL HIGHLIGHTS

- National Champion – Kyokushin Karate (Tamil Nadu State Championship, 2017)
- National Champion – Kyokushin Karate (Delhi National Championship, 2018)
- Bronze Medalist – SGFI Boxing State Tournament, Lucknow (2020)
- Black Belt in Karate – Japan Certified
- Completed Goldman Sachs Governance Analyst Virtual Experience Program; identified legacy password hashing vulnerabilities and proposed modern security hardening recommendations aligned with industry standards.
- Passionate about Cloud-Native Platform Engineering, DevSecOps Automation, and AI-augmented Infrastructure solutions.